



T-Raptor V4 - Technical Documentation

Team Name: One Hertz

Robot Name: T-Raptor V4

Affiliation: IIS L. Cobianchi & Elettra Robotics Lab

1. Introduction and Robot's Initial Overview

The T-Raptor V4 is an autonomous soccer robot developed for robotic soccer competitions. It is the latest evolution in the T-Raptor series. The robot is fully autonomous and uses multiple electronic sensors and mechanical parts to play soccer on a designated field. The team behind this project is called One Hertz. The team consists of two members who divided the tasks to achieve the best efficiency.

Below is the table showing the team members and their specific roles in the project:

Member Name	Core Role	Responsibilities
Nicolò Piana	Hardware Specialist	Physical structure, motor assembly, chassis optimization, and roller calibration.
Luca Bronzi	Software Specialist	Programming, gameplay strategy design, autonomous logic.

We want to give a special acknowledgment to our professors, Sgrò Raimondo and Villa Sergio. They taught us the principles of robot design and programming and guided us throughout the entire creation process.

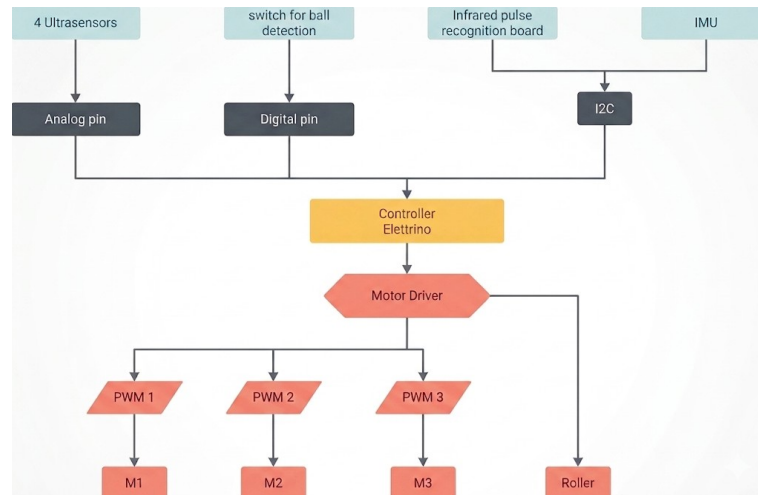


2. Robot's General Logic

The software of T-Raptor V4 is structured to handle multiple real-time situations on the field. The robot behaves completely autonomously based on the values it reads from the environment. The logic can be split into specific behavioral states:

- **Ball Tracking and Attacking:** The primary goal of the robot is to locate the soccer ball. When the ball is detected, the robot moves directly toward it to gain possession and push it toward the opponent's goal.
- **Ball Retention (Dribbling):** When the ball enters the front open area (the mouth) of the robot, a mechanical roller activates automatically. This roller rotates inwards to keep the ball stable inside the robot while moving.
- **Automatic Alignment and Shooting:** When the robot reaches the attacking zone near the opponent's goal, it uses its sensors to face the center of the opponent goal.
- **Defensive Recovery:** If the ball is hidden or out of the sensor range, the robot stops attacking. It immediately drives backward to protect its own goal door and begins scanning the field to find the ball again.
- **Obstacle and Wall Management:** If the robot hits a wall or gets trapped by another robot, an automatic escape routine starts. The robot changes its direction to free itself and re-enters the active game loop.
- **The "Veronica" Maneuver (Anti-Blocking Logic):** When the robot has the ball in its mouth but detects that its path to the opponent's goal is entirely blocked by an adversary, it executes a swift 360-degree rotation. This allows it to circumvent the defender, break the deadlock, and continue the offensive action.

3. Electronic Hardware



The electronic system controls all the components and handles sensor data processing. It consists of a central microcontroller unit and a network of specialized sensors. B

3.1. Main Logic and Power Board



The main controller used on the robot is the Elettrino board, designed by Elettra Robotics Lab.

- It uses a microcontroller compatible with the Arduino Leonardo architecture, making programming fast and reliable.
- It contains an on-board integrated motor driver that handles the power distribution to the motors.

- It features a multifunctional button and multiple status LEDs that help the team perform quick debugging and state switching during matches.

3.1.1. I/O Pinout of Elettrino Board

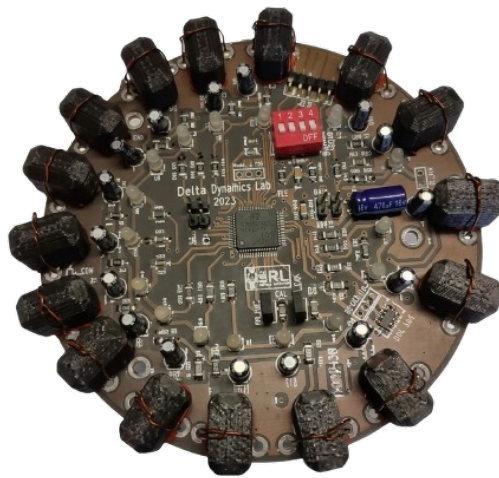
PIN	TYPE	I/O	Function
6	PWM	O	PWM: Speed control for the motor
A5	A	O	H-Bridge Control Pins for Motors
8	D	O	H-Bridge Control Pins for Motors
9	PWM	O	PWM: Speed control for the motor
A3	A	O	H-Bridge Control Pins for Motors
4	D	O	H-Bridge Control Pins for Motors
5	PWM	O	PWM: Speed control for the motor
11	D	O	H-Bridge Control Pins for Motors
10	D	O	H-Bridge Control Pins for Motors
A2	A	I	Ultrasonic (Front): Distance from front wall/objects
A4	A	I	Ultrasonic (Back): Distance from front wall/objects
A1	A	I	Ultrasonic (Left): Distance from front wall/objects
A0	A	I	Ultrasonic (Right): Distance from front wall/objects
7	D	I	Start Button: Start/Stop game logic
13	D	O	Status LED: System initialization signaling
16	D	O	H-Bridge Control Pins for Roller
15	D	O	H-Bridge Control Pins for Roller
1	D	I	Physical contact switch for ball detection

PWM = Pulse Width Modulation | D = Digital | A = analog | I = Input | O = Output

3.2. Sensors and Measuring Units

To perceive the game field accurately, the robot uses three main types of sensory devices:

3.2.1. Infrared (IR) Seeker



The robot finds the ball using an advanced IR Seeker board equipped with 16 infrared sensors. These sensors specifically look for a modulated 40kHz infrared light signal emitted by the official match ball. A dedicated on-board microprocessor continuously reads all 16 channels, calculates the average values, and transmits the exact ball direction and signal strength to the main Elettrino board via the I2C communication protocol

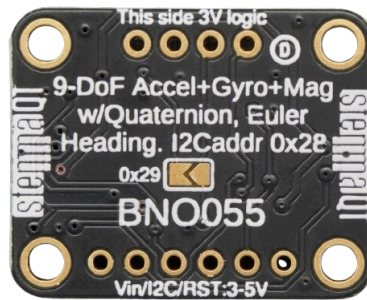
3.2.2. Ultrasonic Distance Sensors

The robot uses four double-capsule ultrasonic sensors positioned on the four sides of the chassis. Double-capsule sensors were chosen because they provide much higher precision than single-capsule models. These sensors serve multiple purposes:

- They measure the distance from the field walls to help the robot know its position.
- They help the robot execute escape routines when it gets stuck near a wall.



3.2.3. Inertial Measurement Unit (IMU)



An IMU sensor (Adafruit BNO055) is used as a highly precise digital compass. The sensor communicates with the main microcontroller via the I2C. When the robot is powered on, the IMU sets the initial direction as zero degrees. During the game, it measures the relative rotation angle. This allows the robot to maintain its absolute orientation, ensuring it always faces the correct direction regardless of collisions.

4. Mechanical and Structural Hardware

The mechanical design focuses on optimization, weight reduction, and modular assembly to ensure easy repairs during intense tournament brackets.

4.1. Chassis Design and Material

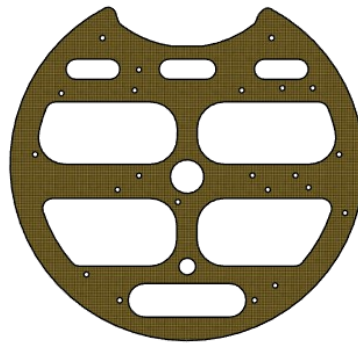
Unlike older versions that utilized fiberglass, the chassis of the T-Raptor V4 is 100% made of 3D printed PLA. The structure is organized into three separate functional levels:

1. **Lower Level:** This base level houses the three movement motors and the IR Seeker. The two front wheels are intentionally positioned at a 140-degree angle from each other. This specific design choice creates the widest possible front intake (mouth) to easily capture and control the ball. Additionally, the structure includes integrated 3D-

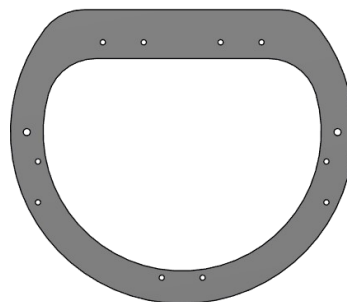
printed suspension parts, protective guards around the wheels, and strategic cutouts to minimize the total weight of the chassis.



2. **Middle Level:** This level holds the main Elettrino logic board, an auxiliary motor driver for the roller, and the main battery pack.



3. **Upper Level:** The top ring acts as a protective shield for the inner electronics and serves as the mounting platform for the upper sensors.



4.2. Motors and Wheels

The locomotion system is designed for omnidirectional movement, allowing the robot to drive in any direction without turning its chassis first.

- **Motors:** The robot uses three Robotzone 12V DC motors. To control the movement direction, the software uses basic mathematical trigonometry to split the target

direction vector into three separate PWM signals, one for each motor.

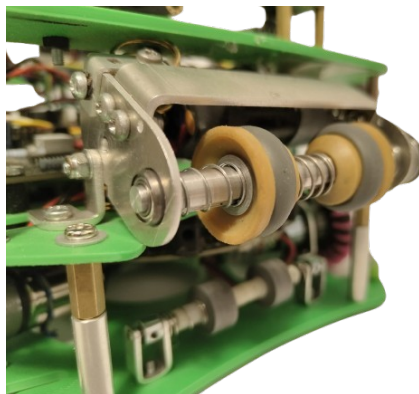


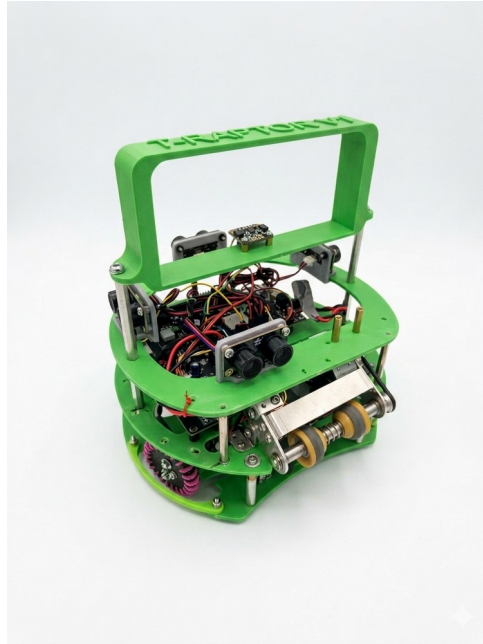
- **Wheels:** The three motors are connected to custom-designed omnidirectional wheels. These wheels are entirely 3D printed and optimized to provide high surface grip on the soccer field carpet while maintaining high durability.



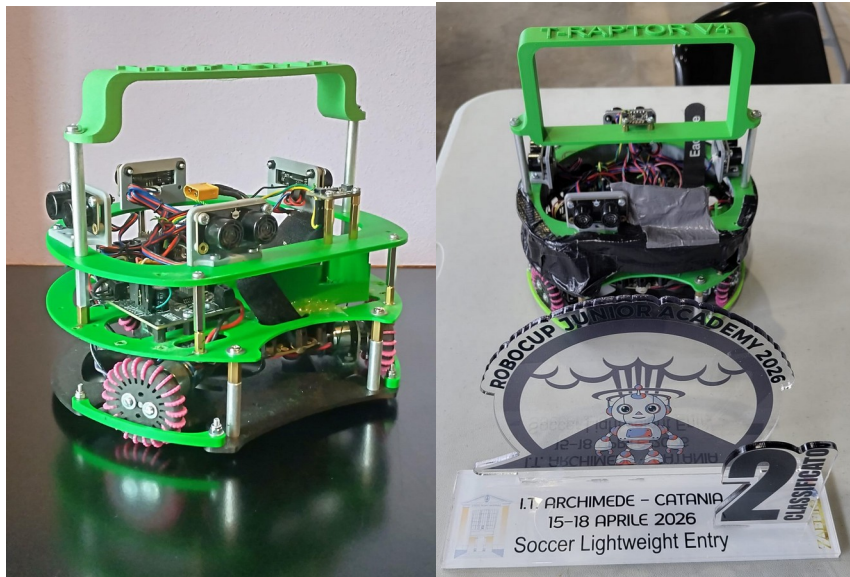
4.3. Roller and Ball Control System

To control the ball efficiently, a custom mechanical roller is mounted at the front of the robot. The roller is driven by an independent motor channel. When a sensor detects the ball, the roller spins inwards to trap the ball against the robot frame, satisfying the active dribbling rules.





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